Introduction

ZZZ is an OEM of mobile self propelled, off-highway, work vehicles. The 4WS Sweeper Vehicle is a proven production unit which is currently operated via hardwired relay logic.

ZZZ has determined that reliance on hard wired relay logic is no longer a viable approach for this product and has contacted STW requesting a hardware and software solution to replace the current relay logic.

Assumptions

The following assumptions are utilized throughout this document::

1. All requirements identified below are driven from the Functional Requirements outlined in the initial Statement of Work document for this project:
   SOW - ZZZ CORP
2. TBD

4WS - Auxiliary Device Functions

FUNCTION 1: Drum Boom Lift Control

Functionality

The Drum Boom Control Module shall control the lifting and lowering of the machine “drum boom” through the use of hydraulic control valves and hardware or CAN switch inputs.

ID	Description
FR-1.1	The control module shall command the machine to perform “Lift” or “Lower” operations based on operator inputs.
FR-1.2	The control module shall prevent simultaneous activation of “Lift” and “Lower” operations.
FR-1.3	The control module shall utilize a “drum boom lift limit” switch to prevent boom lift activation beyond a certain range.
FR-1.4	The control module shall utilize a “drum boom lower limit” switch to prevent boom lower activation beyond a certain range.
FR-1.5	The control module shall invert the default button mapping for lift and lower commands when the inversion flag is set to true.

FUNCTION 2: Sweeper Drum Control

Functionality

The Sweeper Drum Control shall read the operator requested Sweeper Drum commands and convert the requested Sweeper Drum Speed to a PWM output to control the rotational speed of the Sweeper Drum

ID	Description
FR-2.1	The control module shall read the operator requested Sweeper Drum Speed Request from the operator joystick via CAN
FR-2.2	The control module shall read the PTO Engaged condition from the Engine via CAN
FR-2.3	The control module shall apply necessary scaling and clamping conditions to the Drum Speed Request command and bind the speed within minimum and maximum limits.

FR-2.4	The control module shall command the Sweeper Drum Target Speed to zero when the PTO Engaged signal read from the Engine is false.
FR-2.5	The control module shall ramp the Sweeper Drum Target Speed command from the precious value towards the operator requested Sweeper Drum Target Speed
FR-2.6	The control module shall control the speed of the Sweeper Drum and convert the final Sweeper Drum Speed Command into a bounded proportional PWM output that drives the Sweeper Drum coil(s)

4WS - Chassis Functions

FUNCTION 3: Engine Start Control

Functionality

The Engine Start Control Module shall control the engine start operation based on operator ignition input and valid system permissive conditions. The module shall inhibit engine start-up under unsafe conditions.

ID	Description
FR-3.1	The control module shall read the ignition start hardware input and acquire all additional inputs from internal control modules
FR-3.2	The control module shall read in Engine Speed from the Engine (CAN), determine Engine OFF Status is TRUE when engine speed is < 450 RPM, transmit the Engine ON/OFF Status to TBD (led output mounted on joystick, triggered with joystick send msg??)
FR-3.3	The control module shall output the Engine Start Signal to the machine and set to ON only when the ignition start is requested, the Engine is OFF, the joystick is in a valid neutral state, PTO Engaged is FALSE, and operator presence is TRUE.
FR-3.4	The control module shall compute Neutral Safe as TRUE only when the engine is OFF, the joystick is in a valid neutral state, PTO Engaged is FALSE, and Sweeper Drum rotation is less than 10 RPM; otherwise Neutral Safe shall be FALSE
FR-3.5	The control module shall transmit a CAN message containing the Neutral Safe status

FUNCTION 4: Engine Throttle Control

Functionality

The Throttle Control Module shall govern the vehicle's engine speed, measured as crankshaft rpm, by calculating new “Engine Requested Speed” values based on hardware input received from the operator and output via CAN based J1939 speed control messages.

ID	Description
FR-4.1	The control module shall interpret the assigned Joystick buttons for incremental speed increase and decrease as Speed Increment and Speed Decrement Signals
FR-4.2	The control module shall, upon ECU Program Start, command the throttle speed request to be a calibrated minimum setpoint and shall hold this request for 6 seconds before accepting operator adjustments.
FR-4.3	The control module shall increase or decrease, depending on driving input, the engine RPM target at a rate of 100RPM per 100 ms.
FR-4.4	The control module shall ensure mutual exclusivity when handling increment or decrement signals.
FR-4.5	The control module shall output the updated Engine Request Speed to CAN after applying configurable ramping profiles to ensure smooth engine RPM adjustment
FR-4.6	The control module shall set Engine Request Speed and Engine Override Control to zero when the engine is detected to be in OFF for an excess of 6 seconds
FR-4.7	The control module shall populate and transmit all defined TSC1 message signals include the computed Engine Requested Speed.

FUNCTION 5: Vehicle FNR Motion Control

Functionality

The FNR Control Module primarily controls vehicle propulsion in the forward and reverse directions by manipulating the control signal to an electronically controlled, bi-directional, proportional control valve (EFC) responsible for vehicle driving wheel rotation direction and speed.

ID	Description
FR-5.1	The control module shall calculate the Wheel Speed (in MPH) based on hardware feedback provided by a wheel mounted Speed Sensor (Frequency Signal).
FR-5.2	The control module shall read the Main Grip Axis position of the joystick via CAN
FR-5.3	The control module shall Enable EFC (permit movement in any direction) only when: - Engine is ON - ECU Application Program has been running for > 4 seconds
FR-5.4	The control module shall calculate the Target Speed command based on the Main Grip Axis position of the joystick.
FR-5.5	The control module shall utilize the Main Grip Axis position of the joystick to

	determine “position state” of the joystick - Forward, Reverse, or Neutral using predefined ranges / values.
FR-5.6	The control module shall output via CAN message the status of Neutral State to identify when the joystick is in a neutral state.
FR-5.7	The control module shall ramp the target speed command from previous value to new value using pre-configured maximum rates of change
FR-5.8	The control module shall output to dual EFC valves (Forward valve and Reverse valve) based on the target speed command and the state of the joystick
FR-5.9	The control module shall ensure mutual exclusivity between outputs to the forward valve and reverse valve

4WS - Software Helper Functions

FUNCTION 6: PID Output Calculation

Functionality

The PID Calculation Module determines a bounded actuator command by summing proportional, integral, and derivative responses to the difference between setpoint and measured feedback each control cycle.

ID	Description
FR-6.1	The control module shall execute the PID output calculation as a cyclical function at the input parameter control rate.
FR-6.2	The control module shall compute control error as the difference between setpoint and measured feedback.
FR-6.3	The control module shall compute proportional, integral, and derivative terms using input parameter gains and the control period.
FR-6.4	The control module shall sum the proportional, integral, and derivative terms to produce a control output.
FR-6.5	The control module shall limit the control output within input minimum and maximum bounds.

FUNCTION 7: RAMP Value Calculation

Functionality

The Ramp Value Calculation Module shall be used to control how quickly an actuator command is allowed to change over a specified time.

ID	Description
FR-7.1	The control module shall execute the ramp value calculation as a cyclic function using the provided execution period input parameter.
FR-7.2	The control module shall compute an output that transitions linearly from the current output toward the command target value input parameter.
FR-7.3	The control module shall limit the rate of change of the output per execution cycle using the ramp rate and execution period input parameters.
FR-7.4	The control module shall update the ramped output each cycle until the output reaches the commanded target value and shall hold the output constant once reached.
FR-7.5	The control module shall constrain the ramped output within the provided minimum and maximum limit input parameters prior to output.

FUNCTION 8: Toggle Button Behavior

Functionality

The Toggle Button Module shall debounce a momentary button press and toggle a latched output once per press, forcing a safe state when faulted.

ID	Description
FR-8.1	The control module shall execute cyclic toggle logic that measures continuous button press time and shall toggle the latched output state only when the press duration is greater than or equal to the configured debounce time.
FR-8.2	The control module shall allow only one toggle per button press, shall require a button release before accepting a new press, and shall provide the latched state as the toggle output.